LEVEL 8 · PART IV

Kubernetes

8

Goal: Run and troubleshoot applications on Kubernetes with confidence (do this only after Linux, networking, Docker and cloud).

RECOMMENDED TIME
6-10 weeks

TOPICS
19

PRACTICE / CAPSTONE
Move a Node.js API + PostgreSQL + Redis app to a local kind/minikube cluster with Deployments, Services, Ingress, config, probes, HPA, RBAC and NetworkPolicies.

Topics

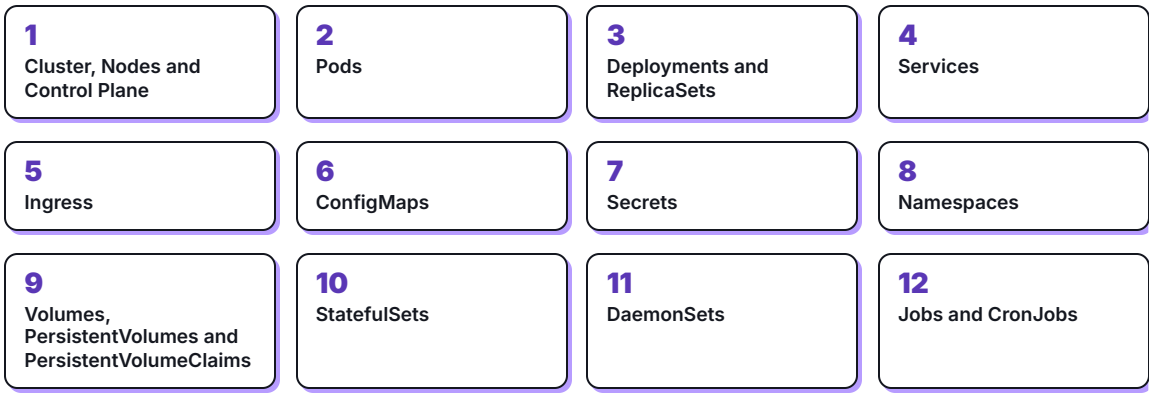
#	Topic	What you will learn
1	Cluster, Nodes and Control Plane **p.1551**	API server, etcd, scheduler, controller manager, kubelet, kube-proxy, kubectl, kind setup
2	Pods **p.1563**	pod spec, multi-container pods, init containers, lifecycle, debugging
3	Deployments and ReplicaSets **p.1577**	declarative scaling, rollouts, rollbacks, strategies
4	Services **p.1589**	ClusterIP, NodePort, LoadBalancer, selectors, Endpoints
5	Ingress **p.1601**	ingress controllers, host/path routing, TLS, cert-manager mention
6	ConfigMaps **p.1615**	env vs volume mounts, immutable config, rollout on change
7	Secrets **p.1625**	types, base64 is not encryption, encryption at rest, external secrets
8	Namespaces **p.1639**	isolation, quotas, LimitRange, multi-tenancy
9	Volumes, PersistentVolumes and PersistentVolumeClaims **p.1653**	StorageClass, dynamic provisioning, access modes, reclaim policy
10	StatefulSets **p.1667**	stable identity, ordered rollout, headless service, volumeClaimTemplates
11	DaemonSets **p.1679**	per-node agents, log shippers, node exporters
12	Jobs and CronJobs **p.1691**	batch, backoffLimit, concurrencyPolicy, TTL

#	Topic	What you will learn
13	Scheduling **p.1703**	nodeSelector, affinity, taints/tolerations, topology spread, priority
14	Resource Requests and Limits **p.1715**	QoS classes, CPU throttling, OOMKilled, right-sizing
15	Liveness and Readiness Probes **p.1727**	startup/liveness/readiness, probe tuning, common outage causes
16	Horizontal Pod Autoscaling **p.1741**	metrics-server, HPA algorithm, behavior, custom metrics intro
17	Service Discovery **p.1753**	CoreDNS, service DNS names, env vars, headless services
18	RBAC **p.1765**	Role, ClusterRole, bindings, ServiceAccounts, kubectl auth can-i
19	NetworkPolicies **p.1779**	default deny, allow rules, CNI support, testing
★	Level capstone project **p.1795**	Move a Node.js API + PostgreSQL + Redis app to a local kind/minikube cluster with Deployments, Services, Ingress, config, probes, HPA, RBAC and NetworkPolicies.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow



13 Scheduling

14 Resource Requests and Limits

15 Liveness and Readiness Probes

16 Horizontal Pod Autoscaling

17 Service Discovery

18 RBAC

19 NetworkPolicies

★ Capstone